# Vikramaditya Singh

vikramaditya-singh.in vikramaditya.bhadoria@gmail.com linkedin.com/in/vikrmadityasngh
github.com/Vikramadtya +91 8383821898

## Profile

Software Engineer with ≈4 years of experience specializing in system design, algorithms, operating systems, and database management. Skilled in developing scalable, efficient software solutions with a strong foundation in data structures and performance optimization. Adept at collaborating with cross-functional teams, communicating complex technical concepts clearly, and adapting to evolving project requirements. Demonstrates a deep technical understanding of full-stack development and cloud platforms, maintaining a strong commitment to high standards of code quality, system reliability, and continuous improvement.

## Experience

**CISCO** **Bangalore, India**
*Software Engineer III* *August 2021 - present*

- Designed, developed, and scaled the backend for the Policy Analyser and Optimiser microservice, enabling the analysis of all Access Policies across FMCs for all tenants on CDO.
- Implemented core analysis logic for rule conflict, object overlap, expired rules, mergeable rules, and never-hit rules detection.
- Created an Access Policy converter microservice, making the Policy Analyser and Optimiser (PAO) version-agnostic, which significantly expanded the customer base.
- Built and maintained CI/CD pipelines in Jenkins, integrating alerting and metrics collection with Datadog.
- Developed a Remediation service to address anomalies identified by the Policy Analyser and Optimiser.
- Created a pub/sub framework for consumers like AI Assistants & AIOPs to subscribe to analysis events.
- Enhanced NAP & IPS policy snapshot creation, cutting operation time by 70% and reducing FMC device configuration deployment time.
- Designed and implemented Zero Trust policy on FMC, including full backend development (API, global search, database, services, validations, reporting, audit, telemetry).
- Developed changes for Zero Trust policy deployment, adding Snort3 configuration creation and enhancing Lina CLI generation with new commands.
- Implemented auto-enrollment of certificates within the Zero Trust policy, significantly enhancing device integration and user experience.
- Improved FMC device listing and management page performance, reducing load time by 96%.
- Added API support for Snort3 rule querying based on the MITRE framework.
- Implemented changes to display rule movement in the audit log, enhancing the user experience.
- Added Elephant Flow Detection to the Access Policy, including backend and deployment changes for Lina and Snort3.
- Integrated PDF reporting, audit logs, delta preview, and telemetry for Elephant Flow Detection & Threat Detection settings.
- Implemented dynamic warning framework for FMC device upgrade flows using React.

**SECURONIX** **Pune, India**
*SDE Intern* *August 2020 - April 2021*

- Developed email microservice and automated incident report generation using Spring Framework and Jasper Reports.
- Created scripts to automate various task.
- Automated report creation for analysis from the data for CTA team.

**CISCO** **Bangalore, India**
*SDE Intern* *May 2020 – July 2020*

- Developed a gRPC detector for Snort 2.9.16 and integrated it with FTD to enable alert generation.
- Worked on the preprocessor framework for scanning the message content transferred via. gRPC for malicious payloads.

## Publication

**Inception Time Model for Structural Damage Detection Using Vibration Measurements** **March 2024**
*In book Fourth Congress on Intelligent Systems (pp.103-122), Springer, Singapore, 31 March 2024*

*DOI: 10.1007/978-981-99-9040-5_7*

## Projects

**Domain Generating Algorithm Detection** | *Python, Keras, Pandas, Google Collab* **code**

- Detected algorithmically generated domains with 96% accuracy and classified them by DGA variant with 87% accuracy, using clustering to identify unknown DGA variants.

**Portfolio Site** | *NextJs, Vercel, Slack* **demo**

- Developed the frontend using Next.js and Tailwind CSS, deployed on Vercel, and integrated with Slack for messaging.
- Utilized Ummami for generating site analytics.

**Tech Blog : NeuralCook** | *NextJs, Vercel, AWS Lambda , FireStore, MailGun, Github Action* **demo** | **code**

- Developed the frontend using Next.js and tailwind leveraged AWS Lambda and Firestore for the backend.
- Automated blog publication using GitHub Actions, utilizing GitHub as a CMS.
- Integrated with Slack for notifications and Mailgun for sending blog email alerts.

## Achievements

**Smart India Hackathon** **2019**

*Finalist for hardware edition*

- Developed a mobile-based irrigation assistant app using Flutter for the frontend and Firebase for backend integration.
- Deployed a LoRaWAN network to extend range and overcome connectivity limitations.

## Awards

**CISCO**

- Connect Everything x4
- Going Above and Beyond x2
- Living Cisco's Principles x1
- Driving the Highest Level of Achievement x1

## Education

**Indian Institute of Information Technology, Guwahati** **2017-2021**

*Bachelor of Technology in Computer Science Engineering* *Cum. GPA: 8.48*

**Modern Delhi Public School, Faridabad** **2015 - 2016**

*12th Science from CBSE Board* *87%*

**Dynasty International School, Faridabad** **2013-2014**

*10th from CBSE Board* *Cum. GPA: 9.8*

## Technical Skills

**Languages**: Java, SQL, Javascript, Python, C, Perl
**Frameworks**: ReactJs, NextJs, Spring, Spring Boot, Hibernate
**Technologies**: Docker, Jenkins, Git, Perforce,Terraform, Lucene
**Databases**: MariaDB, MongoDB, Redis
**Cloud**: AWS
**Developer Tools**: Intellij IDEA, Postman, Data Dog, Draw.io, LocalStack

## Relevant Coursework

- Data Structure & Algorithms
- Computer Network & Security
- Operating System
- RDBMS
- OOP & Design Patterns
- Software Engineering
- Deep Learning